

5th Experiment: Band Structure Simulation for Graphene and CNTs using NanoHub

Fig 1: This figure illustrates the corresponding simulation setup, model, or output in NanoHub. It helps visualize the device structure, selected physics, or simulation results for better understanding.

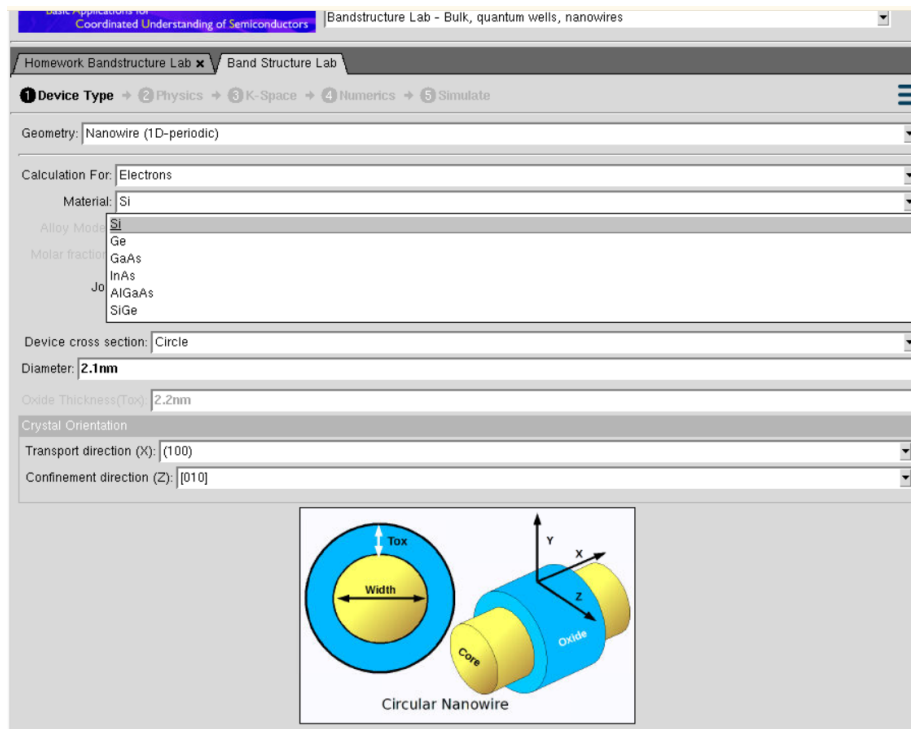


Fig 2: This figure illustrates the corresponding simulation setup, model, or output in NanoHub. It helps visualize the device structure, selected physics, or simulation results for better understanding.

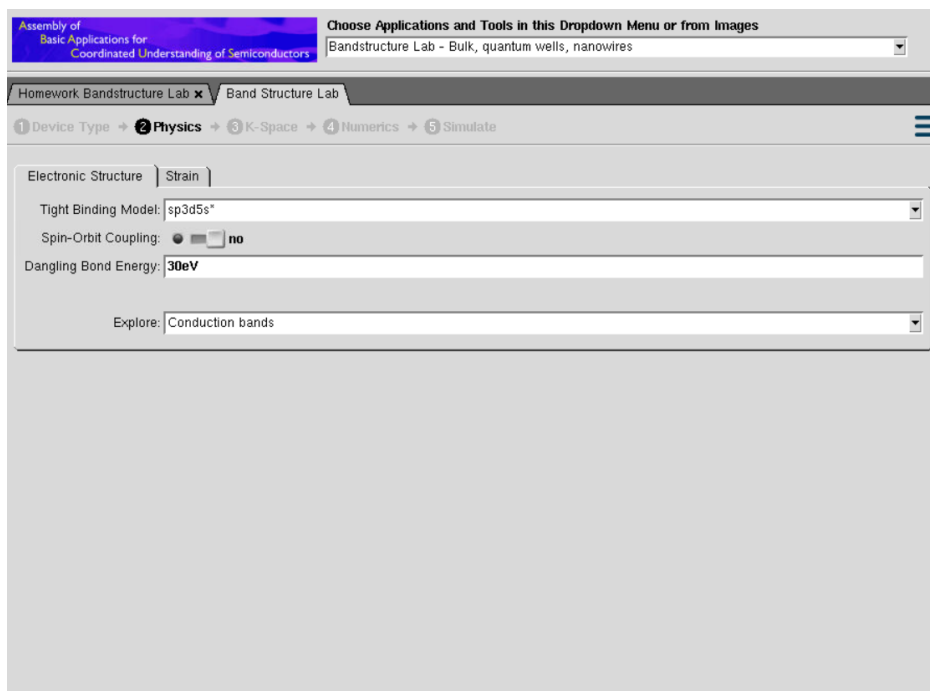
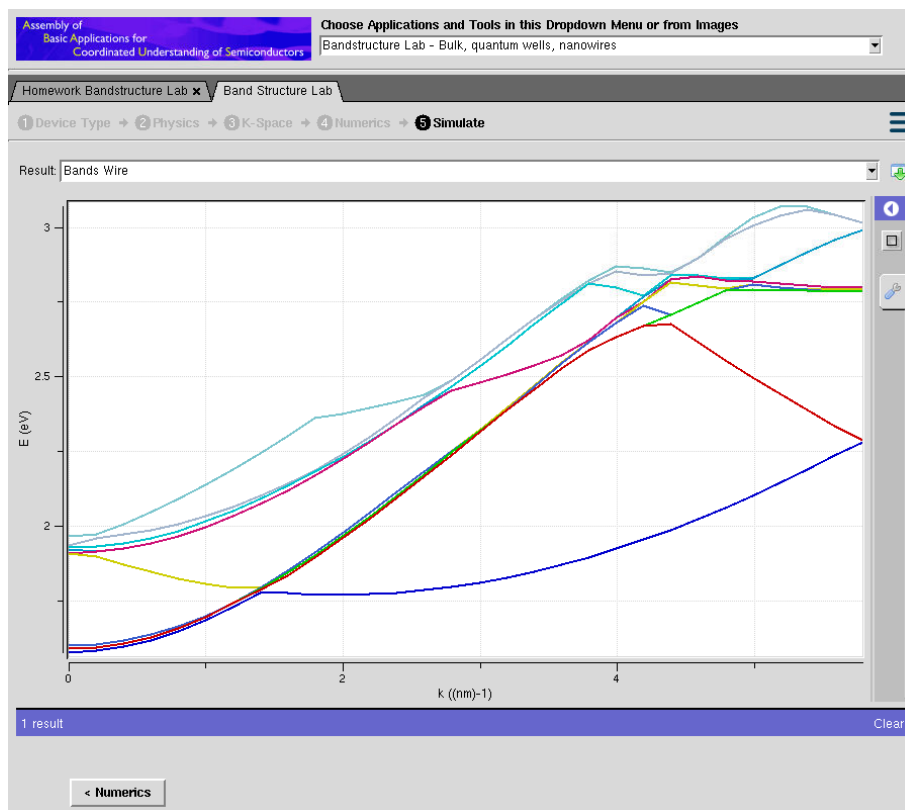


Fig 3: The colored curves represent the variation of electron energy with the wave vector (k), helping to identify the band gap and energy levels.



Case Study 1: This section presents the experiment or case study and its simulation objective. It demonstrates the behaviour and analysis of nanoscale devices using NanoHub.

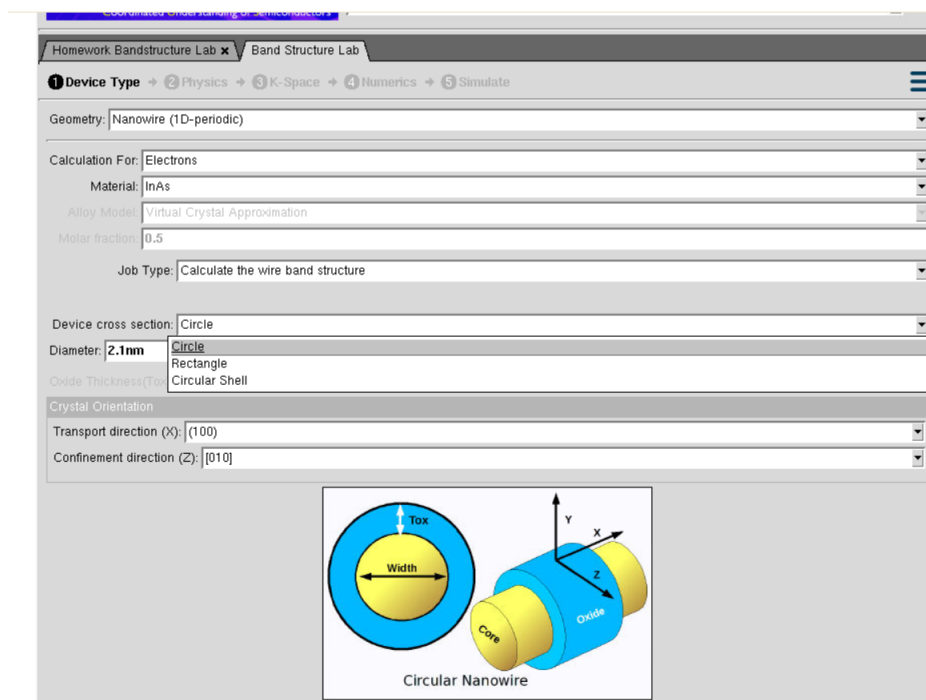
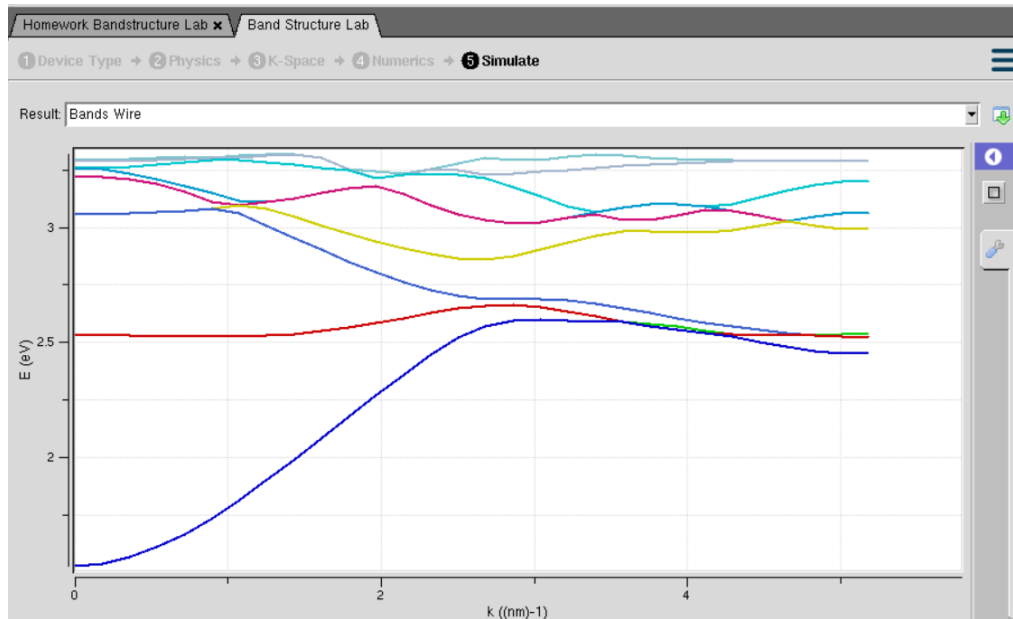


Fig 4: This figure shows the band structure of a semiconductor nanowire simulated using the NanoHub Band Structure Lab. The colored energy bands indicate how electron energy changes with the wave vector (k), revealing the electronic behavior of the material.



Case Study 2: This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

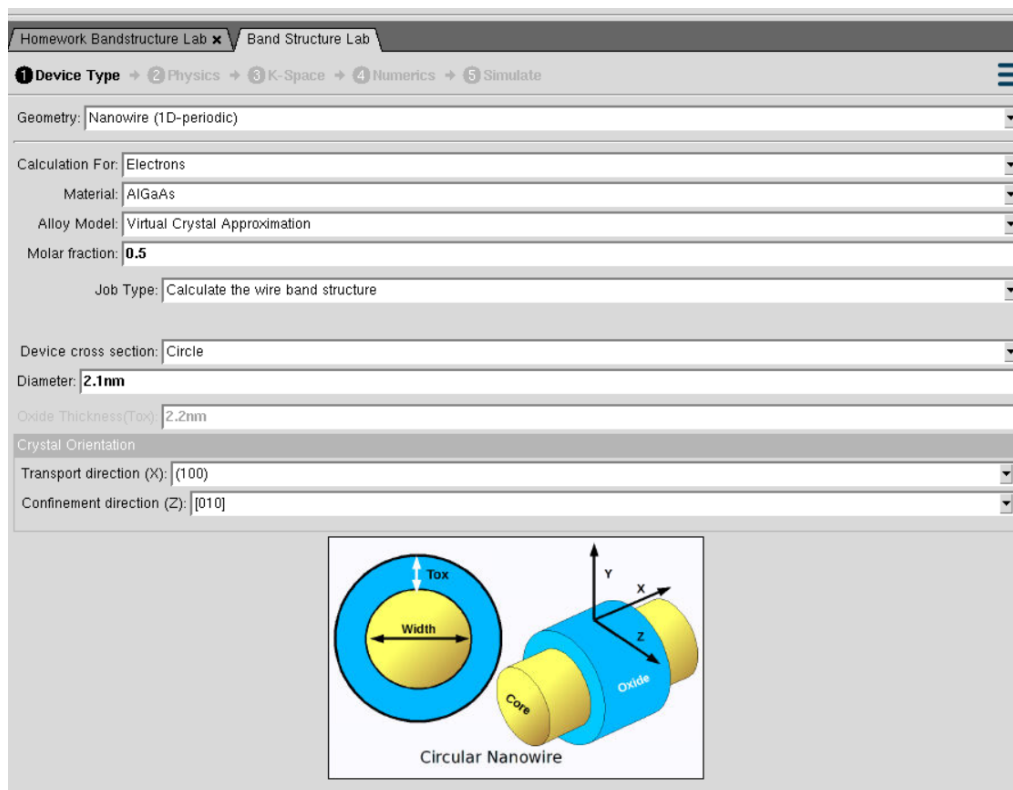
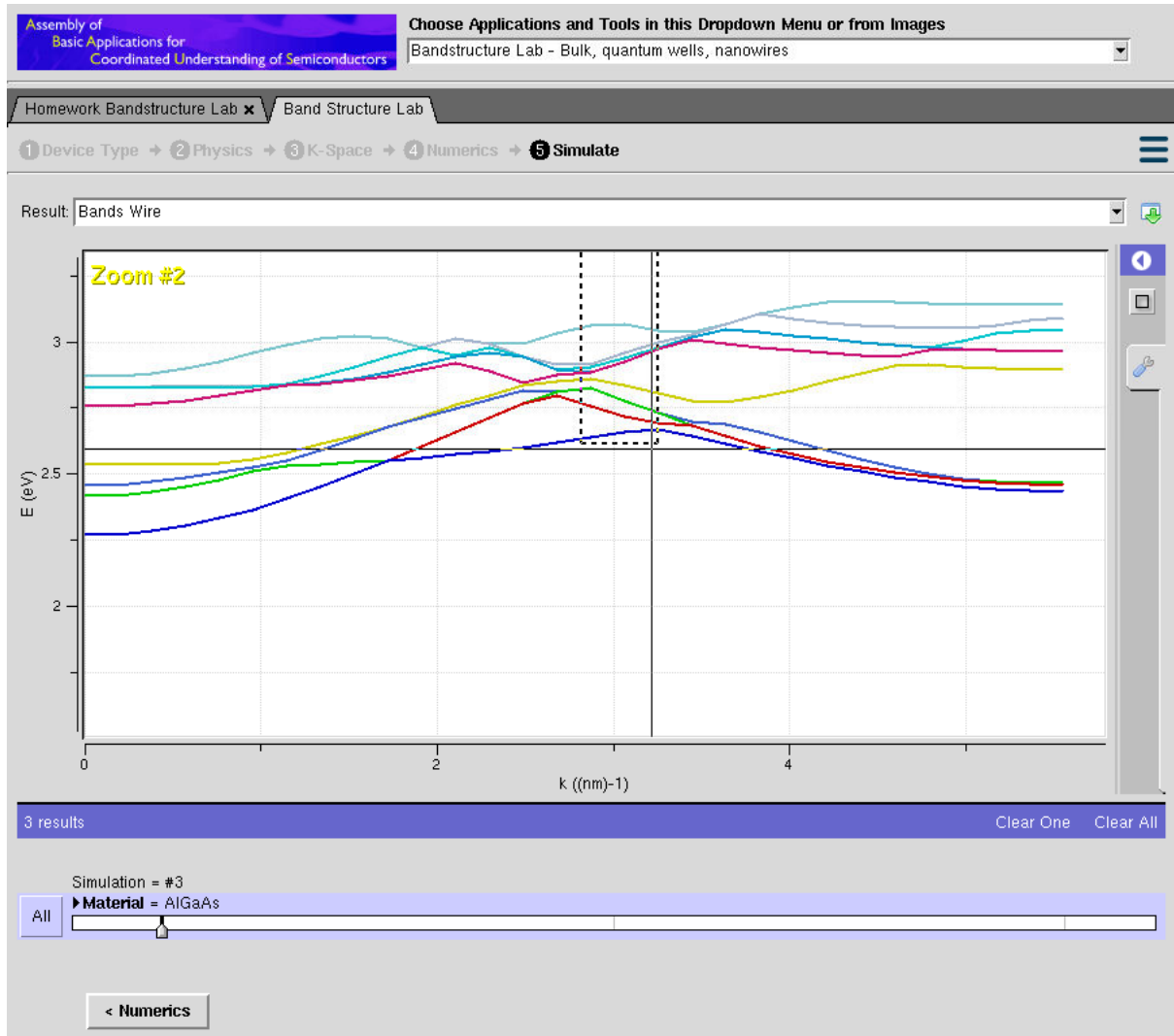


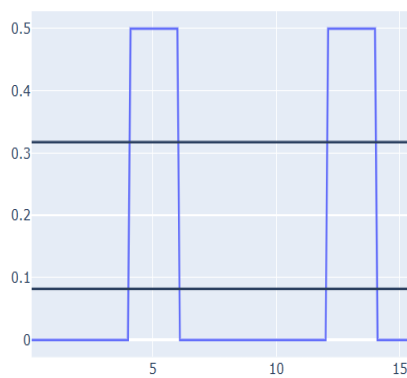
Fig 5: The colored curves represent the allowed energy bands, illustrating how electron energy varies with the wave vector (k).



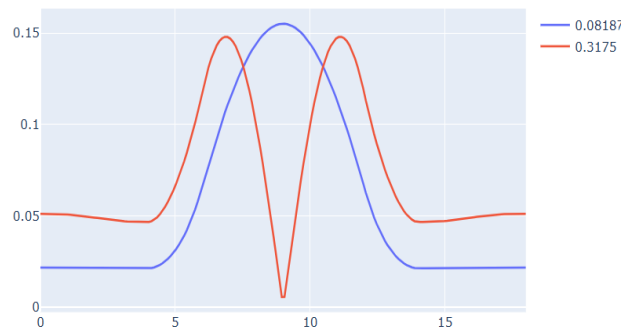
6th Experiment: Demonstration of Quantum Tunneling

This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub

Potential Energy vs Distance



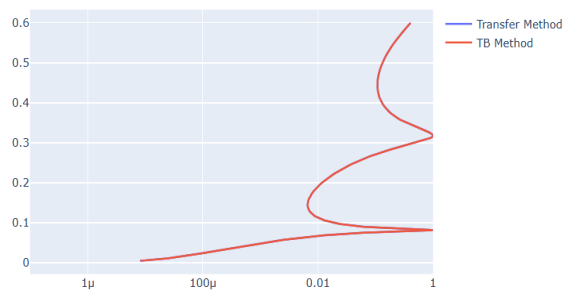
Wave-functions from tight-binding



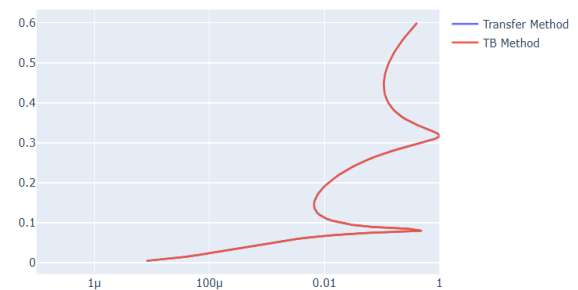
Case Study 1: Analysis the transmission co-efficient Vs Energy characteristics on refined grid

This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

Transmission Coefficient Vs Energy on refined grid



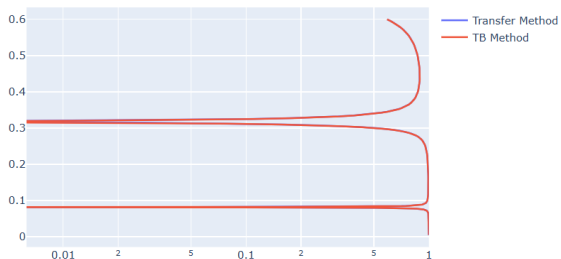
Transmission Coefficient Vs Energy on Homogeneous Grid



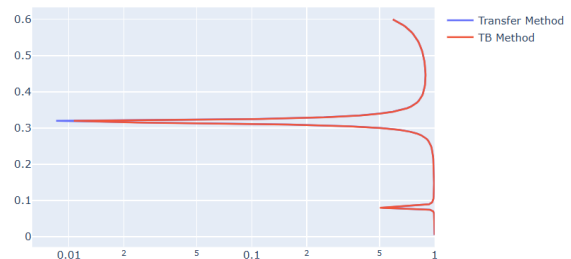
Case Study 2: Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

Reflection Coefficient Vs Energy on refined grid



Reflection Coefficient Vs Energy on Homogeneous Grid



Experiment 7: MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub.

This section presents the experiment or case study and its simulation objective. It demonstrates the behavior and analysis of nanoscale devices using NanoHub.

FETToy

DeviceModelsEnvironment

Model: Single-Gate MOSFET

Gate Insulator Thickness: 1.5nm

Gate Insulator Dielectric Constant: 3.9

Effective Mass Ratio: 0.19

Valley Degeneracy: 2

Floating Boundary Flag: no

Body Thickness: 10nm

Source Doping Density: 1e+20/cm3

Oxide Dielectric Constant

Gate

Electron Masses in Silicon

SiO₂

Source

Channel

Drain

T_{Si}

T_{Ox}

FETToy

DeviceModelsEnvironment

Threshold Voltage: 0.32V

Gate Control Parameter: 0.88

Drain Control Parameter: 0.035

Series Resistance (ohm-um): 0

FETToy

DeviceModelsEnvironment

Ambient Temperature: 300K

Gate Voltage Sweep

Initial Gate Voltage: 0V

Final Gate Voltage: 1V

Gate Voltage Bias Points: 13

Drain Voltage Sweep

Initial Drain Voltage: 0V

Final Drain Voltage: 1V

Drain Voltage Bias Points: 13

Fig 6: The graph shows that the **drain current increases as the gate voltage increases**, indicating normal MOSFET switching behavior. The **red curve** has a higher current than the **blue curve**, showing better performance.

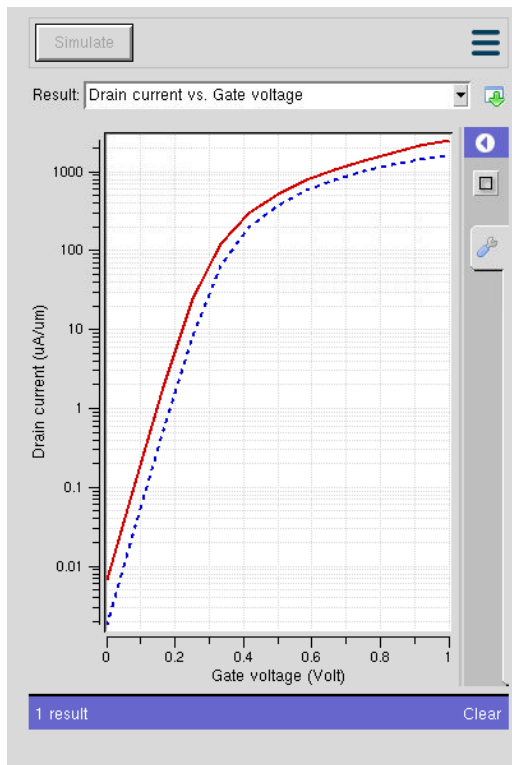


Fig 7: The graph shows that the **drain current increases with drain voltage and then becomes constant (saturation)**. Higher **gate voltage** results in **higher drain current**, indicating normal MOSFET operation.

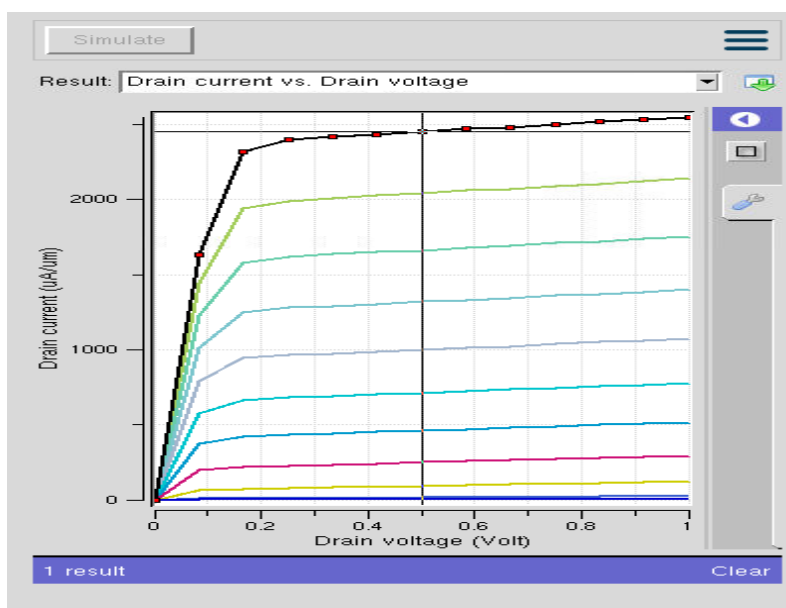


Fig 8: The **gm/I_d ratio decreases as the gate voltage increases**. A **higher gm/I_d at low gate voltage** indicates better transistor efficiency, while a **lower gm/I_d at high gate voltage** indicates stronger inversion operation.

